

Advanced Computer Skills and Hardware

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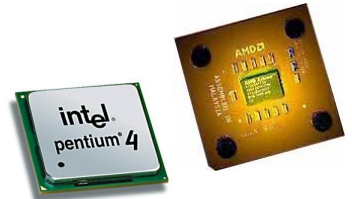
There are 2 types of systems: Intel and AMD

All desktop computers are not created equal. They have significantly different designs. Intel and AMD are the companies that produce the CPU. The CPU of the computer is like the brain, it does all of the processing and is one of the most expensive components.

So what's the difference between Intel and AMD?

AMD – offers the Athlon series processor (Athlon XP3200, Athlon 64)

Intel – offers the Pentium processor (Pentium III, Pentium IV)



Which one is the best?

They are basically equal. It comes down to what you would like to have. Intel boasts to have the faster processor, but AMD is actually the same if not slightly better performance. AMD is also the less expensive choice. It is the best performance for the money. My current recommendation is the AMD systems, but in the future, Intel systems look to be the better choice. We never know what the future may bring in the form of technology.

Motherboard

If the CPU is the brain, then the motherboard would be the nervous system that sends out the signals. A good motherboard will make a fast system. The faster the information can travel the faster the system will run. This is how your other components are connected, such as: CD writers, hard drives, video cards, sound cards and 3 ½ floppy drives. Some motherboards have onboard devices such as sound. Onboard devices used to be something you wanted to avoid but there have been advancements that make onboard sound better than add on sound cards.



Hard Drives

Hard drives are the other large expense. They are necessary for the operation of a computer. They store all the information your computer has, such as: operating systems, programs, pictures and files.



Speed becomes a factor when purchasing and using hard drives. There are 3 different speeds at which hard drives run: 5400, 7200 and 10,000. The faster information can be transferred from your hard drive the quicker your computer will run.

Capacity is always a concern. Most computer users will only need 40GB (gigabytes), which is standard in most machines. If you need to store more information there are many other larger capacities. There is usually a price break between the largest hard drive released and the last months technology. The best method is to divide the number of GB's by the price and find the cheapest price per GB.

Internal hard drives are IDE (Integrated Drive Electronics) and there are 2 different types: **Serial ATA (SATA)** drives are the only drives that operate at 10,000 rpm. Many of the new computers have SATA capability. They are the newer technology and are typically more expensive. They are the best type you can purchase.



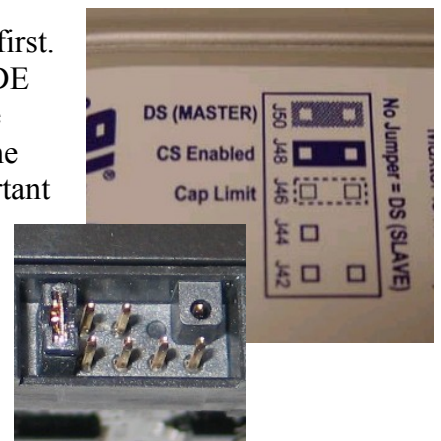
Parallel (ATA = Advanced Technology Attachment) drives are the ones you typically see in stores and are less expensive. I would recommend using the 7200rpm type, since they are the quickest. The difference that's easily seen is the cable size. Serial ATA cables are smaller, whereas the ATA cable is about 6 times as wide.

Something worth considering is that older computers and even newer computers have a 137 GB limit when it comes to hard drives. This can be overcome a number of ways, but is extremely frustrating when you're trying to format a 300GB hard drive and you can only get 137 out of it. These are common ways of overcoming the limit.

- 1) Controller card
- 2) Upgrade Operating system
- 3) Upgrade BIOS
- 4) Purchase a new motherboard
- 5) Use SATA

How to install a new Hard Drive

- 1) Remove your case cover and check to make sure you have room for another hard drive or if replacing; remove the damaged hard drive. The normal space will be a 3 ½ inch slot. Some computers can place a hard drive in a different location but be sure to check your owner's manual first.
- 2) To install a second hard drive you may have to change the IDE cable with the one provided in the hard drive box. The cable must have 2 connectors. The first connector is Master and the second connector (middle) is Slave. This is extremely important because your original hard drive **MUST** be Master.
- 3) To set your original hard drive as master, check the top of the hard drive. It will show a diagram of where to place the jumper. In the example to the right the jumper is placed in the master position. CS stands for cable select, if you connect your hard drive to the Master connector on the IDE cable and the jumper is on CS it will be your Master drive.
- 4) Install the drive and connect the IDE cable and Power.
- 5) Connect the IDE cable to the Motherboard. The boot drive (hard drive that has windows on it) should **ALWAYS** be connected to the primary slot. The primary and secondary slots will be labeled on the motherboard. Secondary is used for CD/DVD drives or additional hard drives.
- 6) The hard drive will then need to be formatted. For a second hard drive start up your computer and use the CD/Floppy that came with your new hard drive.



Formatting will erase all information on the hard drive. Most formatting software is self explanatory. Except when choosing the file system. After formatting is complete, your secondary hard drive is ready for use. Est. time: 15 -60 minutes.

File Systems for Hard Drives

FAT32 – Older file system for storing information. Introduced with Windows 95. Typically only used by system administrators and not usually used by home users.

NTSF – All Windows XP computers will have this file system. It's the most stable and secure system available. Another distinct advantage is that it allows for a single file to exceed 4 gigabytes. This may not seem relevant, but when you have a single file over 4 gigabytes and your computer won't accept it, it becomes a frustrating problem. But don't worry, you can convert a FAT32 drive to NTSF at any time.

Installing Windows

There are several reasons why you may need to install Windows: New computer without an operating system, computer has crashed and needs reinstalled, or upgrading to a new version of Windows.

Boot Disk – a boot disk can be created very easily with a computer running any operating system. A boot disk allows you to start your computer without loading windows. This is a great step if you want to format your hard drive to wipe it clean. To create a boot disk go to "control panel", "add/remove programs". If all else fails you can go to <http://www.bootdisk.com/> and download the boot disk for your operating system. Windows XP does not need a boot disk.

Setting boot order

- 1) The boot order is how your computer looks to startup. If your computer looks at "C:" "Hard Drive" first no matter what you do you won't be able to install windows.
- 2) Immediately when your computer is starting up press F2 or "Delete" to get to your bios menu. Some computers may have you press another button to get to the bios menu. But will have some wording saying "to view bios menu press *" or "setup".
- 3) Once in navigate to "Bios Features Setup".
- 4) Go to Boot Sequence.
- 5) Typical Setup is "A, CDROM, C", when using a boot disk make sure A is first.
- 6) Change options by using the "page up" or "page down" keys on the keyboard.
- 7) Make sure you "SAVE and EXIT".

To format a Hard drive on Windows 95,98,2000,ME, NT

- 1) Insert a boot disk to get to a DOS prompt
- 2) type - format c:
- 3) It will then tell you that everything will be erased and press "Y" for yes.
- 4) Formatting time varies depending on the size of the drive.

Windows XP can be installed through the CD-ROM

- 1) Start your computer and then insert the Windows XP CD into your CD-ROM drive.
- 2) Your computer should automatically detect the CD, and a message "Press any key to boot CD" will be displayed for 5 seconds.
- 3) Press a key and the CD will begin to load files that are needed to begin the installation. This may take a few minutes.
- 4) When asked if you wish to install Windows XP press Enter.
- 5) You will be presented with the End User Licensing Agreement. Press F8 to accept and continue or press F3 to cancel the install.
- 6) Select the partition in which you wish to install Windows XP and press the Enter key. If you wish to delete the current partition or partitions you can do so now. Once you've deleted partitions you will be asked to create a partition. At this point you can choose the size you wish. Most people will elect to create one partition that fills the entire drive. If you decide you would like multiple partitions then create a partition size that will be suitable for XP and your installed apps as well as room for cache files etc. 10GB is a nice size for a full XP install leaving plenty of room for applications etc.
- 7) The next screen asks if you wish to use the NTFS file system. This is the preferred file system; however, if you choose to use FAT32, you will not have all the security and stability features of Windows XP.
- 8) If you previously had an Operating System installed you will be given a choice of Format options. Select the Format option of your choice. When asked to start the format, press the "F" key. The formatting process may take quite a bit of time.
- 9) Setup will begin an automated loading of files which will take several minutes.
***Note - If you are clean installing using an XP Upgrade CD you will be asked for a qualifying product at this point. Just pop the CD for your older version of Windows into the drive while the XP setup completes an authenticity check. Following this you will be prompted for the XP CD and setup will continue.
- 10) After this is complete the computer will restart. Leave the XP CD in the drive but this time **DO NOT** press any key when the message "Press any key to boot CD" is displayed. In 5 seconds setup will continue.
- 11) The Windows XP Setup wizard will lead you through the process of gathering information about you and your computer.
- 12) If you are connected to the Internet XP will prompt you to "Activate". Do not activate at this time. You will be prompted periodically to activate but you have 30 days to do so. Remember though that at the end of 30 days, if you have still not activated you will no longer be able to access the Desktop.
- 13) Following the Activation Screen you will be given the opportunity to Register Windows XP. Registration is optional and choosing not to register will not have any negative effect on your system.
- 14) You will be required to enter your account password to gain access to the Windows XP Desktop. (If you do not enter a password and simply click "OK" your system will not ask you to login and will boot to the desktop automatically.)

Software Drivers

A driver is a link between your hardware and the software that runs it. A proper driver is necessary for the hardware to function properly. Always keep the installation discs that come with your hardware. If for some reason you lost your install disc or a device stops functioning, go to the manufacturer's website and download it. If all else fails try searching the internet. <http://www.driverfiles.net/> is a collection of common drivers.

Device Manager

The device manager shows all of your devices (hardware). If a device is not functioning properly you will need to access your device manager. Go to control panel and then open "SYSTEM", there will either be a tab that says device manager or you will have to navigate through the tabs to find "Device Manager". Once in the device manager you can press the pluses next to the different categories. If at any time you see a yellow exclamation mark, that device is not working properly. To check what is wrong, right click on the device and go to "PROPERTIES".

Task Manager

The task manager can be accessed by pressing "CTRL+ALT+DEL" once. (For windows XP you may then have to click the "task manager" button). The task manager allows you to see everything your computer is currently running. If the computer freezes and you can still access the task manager you can end any "Not responding" tasks. A "Not Responding" task is frozen and will more than likely not recover, ending the task will allow you to continue working on your computer. By pressing "CTRL+ALT+DEL" a second time your computer will be restarted or you can choose "Shut Down". The average computer uses around 40 processes.

Add/Remove Programs:

Installing programs is very easy and most programs automatically guide you through the install process. Uninstalling can sometimes be more difficult. Open control panel and double click "Add/Remove Programs". Not only will all the installed programs be displayed but in windows XP it will show when it was installed and the frequency of use. If the program you want to remove is not in the list there is only one option left. You must delete the folder that has the program itself, this is not a recommended option, but if no other choice exists it can be used. An example would be spyware or adware programs that don't want to leave your system and can't be uninstalled.

Startup (msconfig):

When a computer starts up many "hidden" programs run, this in turn slows down a computer and makes it more likely to crash. Many of these programs can be disabled with no problems incurred; in fact the system will perform better. Unfortunately Windows 2000, 95, and NT do not have the msconfig utility. But a similar utility can be downloaded at: <http://www.mlin.net/StartupCPL.shtml> . For others uses simply click

the start button, click run, and type “msconfig” and a utility box will appear. Click on the tab that says “startup”, you will then be provided with a list of all the programs that start and run every time the computer boots up. But which are good and which are bad/not necessary. <http://www.processlibrary.com/> gives a searchable list and directory of many common startup items. It also lists viruses, spyware and dangerous items. You can always uncheck an item you’re unsure about and restart and if everything seems to function normally, you’re fine.

Firewalls:

Firewalls protect you from malicious attempts by outside sources. Think of it as a wall of fire around your computer that is very difficult to penetrate. Windows XP has a built in firewall. Usually it is automatically turned on. To access the XP firewall, right click on “My Network” and choose properties. Then right click on the internet connection you want to protect, and click properties. (ex. Local Area Connection). In the new window click the “Advanced” tab. Then just checkmark the first choice for protect, or uncheck to disable it. Why would you want to disable a firewall? Sometimes firewalls can interfere with other programs and may need to be permanently or temporarily turned off. If you do not have windows XP there is a free firewall program called ZoneAlarm. It’s much better than the built in firewall of Windows XP. Many XP users disable the built in firewall and use the free ZoneAlarm.

Free Firewall:

<http://www.zonelabs.com/store/content/company/products/znalm/freeDownload.jsp>

Defragment and ScanDisk

You can use Disk Defragmenter to rearrange all of your built-up file fragments into complete file blocks, which will speed up file access in your hard drive.

- 1) Click **Start > Programs > Accessories > System Tools > Disk Defragmenter**
- 2) Click Analyze to determine if you should defragment the hard drive.
- 3) Click Defragment to start defragment of the selected hard drive. Depending on the fragmentation of your hard drive, this process can take several minutes to a few hours.
- 4) If possible defragment your hard drive once a month.

ScanDisk checks your hard drive for any errors. To keep your hard drive free of physical problems such as scratches or demagnetized area’s, occasionally run ScanDisk.

- 1) Click **Start > Programs > Accessories > System Tools > ScanDisk**
- 2) Select the drive you wish to scan.
- 3) Select either Standard or Thorough, make sure the check box “Automatically Fix errors” is check marked.
- 4) Let the program run. Don’t use your computer for anything else, disable a screensaver if you have one, otherwise the scan will restart.

Viruses, Spyware, Adware, Trojans and Worms

What to do when an infection occurs? The 2 main virus scan programs are Norton and McAfee. Main being that you have to pay for them and pay for updates. There are plenty of free alternatives that are just as effective and sometimes pick up virus's the main programs miss.

Free Antivirus Programs:

Grisoft: <http://free.grisoft.com>

(Scroll down and click Download your AVG free)

AntiVir: <http://www.free-av.com>

Avast: <http://www.avast.com/eng/download.html>

(Free edition for home users)

Trend Micro: <http://housecall.trendmicro.com>

(Online scan, best with a high speed internet connection, extremely powerful and will provide instructions for removal of dangerous viruses)

Free Spyware/Adware Removal:

Most spyware is frustrating and hard to get rid of. With the following scans you can eliminate most threats. Spyware blocking is next to impossible, the only step is scanning often and removing what is on your computer.

Ad-Aware: <http://www.lavasoftusa.com/>

(One of the best out, but will not catch everything)

Spybot Search and Destroy: <http://www.safer-networking.org>

(Provides some Immunizing features which will help block some spyware from getting on your computer)

Microsoft Anti-Spyware: www.microsoft.com (go to the downloads section)

(It will scan your computer daily at a selected time automatically)

Safe Mode:

Safe mode is used when your computer is badly infected and needs protected from itself. In safe mode you can run all the virus scans and spyware scans and removal will be easier. Safe mode does not allow an internet connection which is perfect because most spyware programs require the internet to reinstall themselves.

To use the F8 key to start Windows 98/Me/2000/XP in Safe mode

1. Restart the computer.
2. As the computer restarts, press and hold down or Tap the **F8** key until the Windows 98/Me/2000/xpStartup menu appears.
3. When the Startup Menu appears, select Safe Mode and press Enter.